

采样 (sampler)

EDM里面的欧拉采样

```
def sample_euler(model, sigmas: list, N):
    timesteps = sigmas # EDM里面用的
    x = torch.zeros() + torch.randn(1, 3, 64, 64) * sigmas[0] # 采样纯噪声x_0出来
    for i in range(N):
        x_pred = model(x, sigmas[i]) # 推理, 预测干净图x_pred
        d = (x - x_pred) / sigmas[i]
        # 也就是  $dx / dt = (x_{pred} - x) / (timesteps[N-1] - timesteps[i])$ 
        #  $= (x_{pred} - x) / (sigmas[N-1] - sigmas[i])$ 
        #  $= (x_{pred} - x) / (0 - sigmas[i])$ 
        #  $= (x - x_{pred}) / sigmas[i]$ 
        x_i = x_i + d * dt
        # 相当于  $x_{i+1} = x_i + dx / dt * dt$ 
        #  $= x_i + dx$ 
```

sigmas噪声的设定 (scheduler)

comfy.model_base.model_sampling.ModelSamplingDiscrete (DDPM)

推理

封装过程: BaseModel.apply_model (UNet) - ModelPatcher、CFGGuider、comfy.samplers.KSamplerX0Inpaint

UNet的输入输出的预处理、后处理操作: comfy.model_sampling.EPS